

Spanish-Mapudungun Translation using Transfer Learning for Low-resource Languages

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Abstract—This work addresses Neural Machine Translation (NMT) for Spanish–Mapudungun, an endangered and digitally under-resourced indigenous language of South America with polysynthetic and agglutinative morphology. We explore transfer learning as a strategy to mitigate data scarcity, initializing Spanish–Mapudungun models from high-resource pairs (Spanish–English and Spanish–Finnish). Using a standardized 10,000-pair corpus, we show that transfer learning markedly improves translation quality in both directions over models trained from scratch. We further analyze whether the linguistic characteristics of the parent language—particularly morphological similarity—affect transfer effectiveness. Results indicate that both parent models achieve similar gains, suggesting that vocabulary coverage and tokenization dominate over morphological resemblance. These findings demonstrate the utility of transfer learning for endangered languages and underscore the need for linguistically informed methods and community involvement in revitalization efforts.

Index Terms—neural machine translation, transfer learning, low-resource languages

I. INTRODUCTION

Mapudungun, the ancestral language of the Mapuche people in Chile and Argentina, is a polysynthetic and agglutinative language with complex grammar and a lexicon deeply tied to cultural identity. Despite this significance, it is classified as endangered: while around 10% of Chile’s population are Mapuche descendants, fewer than 1% actively speak the language due to the dominance of Spanish driven by historical and sociopolitical factors [1].

Developing translation systems between Spanish and Mapudungun is critical for language preservation and revitalization. Such systems can support access to education, healthcare, and public services, while promoting cultural continuity and inclusion by bridging the linguistic divide with the Spanish-speaking majority.

However, translating Spanish into Mapudungun poses major challenges. Mapudungun is a low-resource language with scarce corpora and limited computational research, leaving few effective translation tools. Recent advances in Neural Machine Translation (NMT) and transfer learning provide a way to address these constraints [2].

This study investigates transfer learning for Spanish–Mapudungun translation to answer:

- 1) Can transfer learning improve translation in both directions?
- 2) Does morphological similarity between parent and target languages—particularly their shared agglutinative structure—affect transfer effectiveness?

We compare baseline NMT models trained from scratch with models initialized from Spanish–Finnish and Spanish–English parent models, and evaluate custom tokenization strategies tailored to Mapudungun morphology. Beyond building a translation model, we aim to support language preservation and community engagement by releasing a web application for early feedback from Mapuche speakers.

II. BACKGROUND

A. Neural Machine Translation

Neural Machine Translation (NMT) models translation as a sequence-to-sequence learning problem using deep neural networks, replacing rule-based and statistical approaches that rely on handcrafted features and co-occurrence statistics. Given a parallel corpus, NMT learns distributed representations of language in a data-driven manner, making it the dominant paradigm for machine translation [3].

Modern systems are based on an encoder–decoder architecture with attention. The encoder maps a source sentence $\mathbf{x} = (x_1, \dots, x_n)$ into latent vectors $\mathbf{h} = (h_1, \dots, h_n)$, and the decoder generates the target sequence $\mathbf{y} = (y_1, \dots, y_m)$ by modeling $P(\mathbf{y}|\mathbf{x})$. Attention mechanisms [4] allow the decoder to focus on relevant source tokens, overcoming the limitations of fixed-length context.

Compared to traditional MT, NMT enables end-to-end learning, better captures long-range syntactic and semantic dependencies, and adapts readily to new domains through fine-tuning.

a) Challenges of NMT for Low-resource Languages:

NMT systems struggle with low-resource languages because small parallel corpora severely degrade performance and often cause models to overfit by memorizing training examples rather than generalizing. Limited vocabulary coverage further complicates learning, as standard subword tokenizers—especially those pretrained on unrelated languages—fail to capture morphemes in highly inflected languages. For Mapudungun, additional challenges include significant regional

variation in lexicon and grammar, which makes consistent mappings difficult to learn, and substantial typological and syntactic divergence from high-resource languages, which reduces the effectiveness of transfer from multilingual models [5].

B. Transfer Learning in NMT for Low Resource Languages

Transfer learning mitigates data scarcity by reusing knowledge from a high-resource task $(\mathcal{T}_s, \mathcal{D}_s)$ to initialize a low-resource model $(\mathcal{T}_t, \mathcal{D}_t)$ where $|\mathcal{D}_t| \ll |\mathcal{D}_s|$. Zoph et al. [6] formalized this approach for NMT: a parent model trained on a large corpus (e.g., Spanish–English) is fine-tuned on the smaller target pair (e.g., Spanish–Mapudungun), improving BLEU and training stability.

a) Language Similarity as a Transfer Factor: While transfer benefits often increase with linguistic proximity between parent and target languages [7], recent work shows that tokenization and vocabulary overlap can outweigh structural similarity, particularly for morphologically rich languages.

C. Mapudungun: A Case Study in Low-Resource NMT

Mapudungun (ISO 639-2: *arn*) illustrates many of the structural and resource-related hurdles in low-resource NMT:

a) Linguistic Complexity: Mapudungun is a polysynthetic, agglutinative language from the Araucanian family, with complex verbal morphology and a productive affixation system [2]. A single word in Mapudungun can encode the semantic content of an entire clause in Spanish. For instance, the word *Tüñkülepürakatun* (“now I am calmer”) combines multiple morphemes: *tüñküle* (calm) + *püra* (intensifier) + *katun* (verb inflection).

b) Resource Constraints and Implications: Most Mapudungun data exists in spoken form, with limited standardized orthography and few digitized corpora. The largest available dataset is the AVENUE corpus, product of a collaboration between Carnegie Mellon University and the Chilean Ministry of Education. It comprises approximately 260,000 annotated phrases, but even this must be heavily cleaned and standardized for NMT training [1].

Given the scarcity and linguistic uniqueness of Mapudungun, effective NMT requires custom subword tokenization—such as SentencePiece with BPE—trained on monolingual Mapudungun text to capture meaningful morphemes. Parent model selection can consider morphosyntactic similarity, for example Spanish–Finnish, although our experiments suggest that token-level factors may be more influential. Equally important is community-informed data validation to ensure linguistic and cultural appropriateness, as automated metrics like BLEU cannot fully capture semantic adequacy or respectful usage. These insights position Mapudungun as a prototypical case for advancing low-resource NMT and for designing inclusive NLP technologies for endangered languages.

III. RELATED WORK

Translation between Spanish and Mapudungun has progressed from traditional linguistic documentation to compu-

tational approaches. Early resources such as bilingual dictionaries and grammars provided essential descriptions but are insufficient for large-scale translation. Initial rule-based systems attempted direct sentence conversion but were limited by their manual design and inability to handle Mapudungun’s complex polysynthetic structure [8]. More recently, community-driven initiatives, often in the form of apps or basic tools, have supported language preservation, yet these efforts remain constrained by scarce linguistic data and limited computational resources.

A. Translation for Low-resource Languages

Neural Machine Translation (NMT) has advanced low-resource translation, though data scarcity and linguistic diversity remain major obstacles. Two key strategies are widely used: *transfer learning*, where models pre-trained on high-resource pairs are fine-tuned on low-resource data [6], and *back-translation* [9], which augments training sets with pseudo-parallel data generated from monolingual texts. Studies (e.g., German–Bavarian [10], Egyptian–MSA) show that language similarity can enhance transfer, while synthetic data significantly improves performance.

Multilingual pre-trained models such as mBERT and XLM-R extend these gains through cross-lingual transfer, with efficient adaptation methods including LoRA and federated prompt tuning that reduce data and privacy constraints [11].

For Spanish–Mapudungun, Duan et al. [1] trained a Transformer model from scratch using the parallel corpora of 260,000 pairs from the AVENUE project. Without transfer learning, performance was limited by the corpus size and the morphological complexity of Mapudungun.

B. Chilean Initiatives

In Chile, Pendas et al. [12] applied active learning to reduce the annotation burden on native speakers when selecting training examples for Spanish–Mapudungun translation. While effective for data collection, these efforts rely on standard models and do not address architectural adaptations for Mapudungun’s complexity. Significant gaps remain in building scalable systems that combine advanced NMT methods with community-driven evaluation. Our work extends these approaches through transfer learning and a web platform designed to incorporate user feedback, targeting both technical and cultural aspects of language revitalization.

IV. APPROACH

This study investigates transfer learning to improve Neural Machine Translation (NMT) for the Spanish–Mapudungun (*es*↔*arn*) pair under severe data scarcity. Mapudungun’s polysynthetic, agglutinative morphology and limited digital resources pose challenges for conventional NMT. We evaluate whether transfer learning can enhance translation quality and examine the influence of morphological similarity between parent and target languages.

We adopt a two-stage experimental framework. First, we train a Transformer-based baseline from scratch on a subset

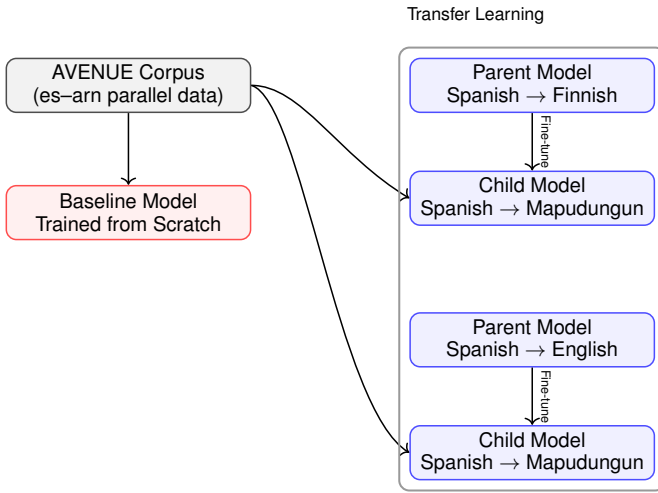


Fig. 1. Overview of the transfer learning approach. Pre-trained parent models on Spanish-Finnish and Spanish-English are fine-tuned using the Spanish-Mapudungun AVENUE corpus. Their performance is compared against a baseline model trained from scratch.

of the AVENUE corpus [2] and compare it to models fine-tuned from multilingual Helsinki-NLP Transformers. Second, we evaluate transfer models initialized from two parent pairs: Spanish-Finnish ($es \leftrightarrow fi$), where Finnish shares agglutinative morphology with Mapudungun, and Spanish-English ($es \leftrightarrow en$), a morphologically distant language. This setup isolates the effect of morphological similarity on transfer performance. Finnish was selected as an agglutinative parent language due to its structural parallels with Mapudungun (complex suffixation, compounding, morpheme fusion), potentially enhancing inductive transfer. English, lacking such morphology, serves as a control to disentangle grammatical effects from factors like vocabulary overlap and tokenization. Additional agglutinative languages were also considered.

All experiments use the MarianMT Transformer architecture [13], optimized for low-resource scenarios, and evaluate both directions ($es \rightarrow arn$ and $arn \rightarrow es$). Models are assessed with BLEU [14] and chrF [15] under comparable computational settings. By combining typological considerations with transfer learning, this study provides insights into cross-lingual adaptation and guidelines for NMT in endangered languages.

A. Dataset

As mentioned previously, the dataset used for training is a curated subset of the Mapudungun-Spanish parallel AVENUE corpus [2]. The full corpus comprises approximately 260,000 phrase pairs, each consisting of a transcription of a spoken Mapudungun utterance and its corresponding human-generated Spanish translation. These utterances were collected from real conversational settings, making the corpus a linguistically rich but structurally noisy resource.

For quality reasons, a subset of 10,000 sentence pairs was sampled for training, along with validation and test sets comprising 1,250 pairs each. This fixed-size sampling strategy

is maintained consistently across both baseline and transfer learning experiments to ensure comparability.

a) Data Processing: Processing the parallel corpus is essential to mitigate inconsistencies and ensure high-quality input for model training. One of the primary issues is the lack of uniform formatting across the dataset, a consequence of contributions from multiple annotators with varying transcription practices. Mapudungun and Spanish segments are typically prefixed by markers such as “M:” and “C:”, but these notations are inconsistently applied. Some entries place both source and target on a single line, while others vary in delimiter syntax complicating automated extraction.

To mitigate these inconsistencies, a custom processing pipeline was developed using regular expression matching to reliably identify and align sentence pairs. These expressions were crafted to detect and isolate Mapudungun and Spanish segments, regardless of inconsistent prefix usage (e.g., “M:”, “C:”, “C.”). This step was critical for enforcing a consistent input-output structure for translation pairs. Each extracted pair was normalized to follow a uniform format, with Mapudungun phrases always positioned as the source or target, depending on the translation direction, and Spanish phrases as their corresponding counterpart. Additional cleaning procedures were applied to remove annotation artifacts (e.g., “ Noise_i ”, “ Noise_i ”), standardize punctuation, and correct spacing errors. Diacritics and capitalization were normalized to ensure compatibility with downstream tokenization.

An illustrative example of a raw aligned phrase pair prior to tokenization is:

M: fey manchana ka fey ka ti.
C: la manzana tambi  n dicen.

This example highlights several common features of the dataset, including informal conversational structure, frequent conjunctions and repetitions, and orthographic inconsistencies (e.g., “tambi  n” with a grave accent).

B. Baseline Model Training

To quantify transfer learning gains, we first train a baseline NMT model from scratch on the Spanish-Mapudungun corpus. A standard Transformer sequence-to-sequence model is trained with cross-entropy loss and early stopping on validation BLEU. Preprocessing (sentence extraction, normalization, and noise removal) is applied identically to baseline and transfer setups to ensure comparability.

a) Model Architecture: The baseline follows the Transformer architecture [16] used by Duan et al. [1]. It employs a 5-layer encoder-decoder with multi-head self-attention (8 heads) and position-wise feedforward networks (2048 hidden units). Tokens are embedded in a 512-dimensional space with positional encodings. Regularization includes dropout (0.4) and label smoothing (0.1). Models are trained with Adam, which is robust for low-resource regimes.

b) Limitations of the Baseline Model: The baseline Transformer trained from scratch suffers from the typical constraints of low-resource NMT. Training on 10,000 sentence pairs limits generalization and leads to overfitting, particularly

on rare constructions and domain-specific vocabulary. Mapudungun’s agglutinative, polysynthetic morphology generates a vast number of surface forms, causing poor vocabulary coverage and making generation more challenging than translation into Spanish. Duan et al. [1] report BLEU scores of 6.26 (arn→es) and 1.09 (es→arn), highlighting this asymmetry. Additionally, generic tokenizers inadequately segment complex word forms, further exacerbating sparsity.

C. Transfer Learning Implementation

To address the limitations of baseline training, we fine-tune pre-trained MarianMT models from the Helsinki-NLP Opus-MT collection [17]. Spanish–Mapudungun child models are initialized from Spanish–Finnish and Spanish–English parents to examine how linguistic similarity, particularly agglutinative morphology, affects transfer. MarianMT [13] provides an optimized Transformer architecture with multilingual vocabularies, SentencePiece tokenization, and efficient batching, suitable for low-resource adaptation.

Fine-tuning uses the same training pairs as the baseline, with a fixed learning rate, weight decay, and small batch size. To reduce vocabulary mismatches, tokenizers are extended with Mapudungun-specific subwords, allowing better encoding of its rich morphology. This process transfers general translation knowledge from the parent models and adapts it to the low-resource domain. The following sections compare these fine-tuned models with the baseline to quantify the benefits of transfer learning.

a) Parent Model Selection: Transfer learning uses pre-trained from the Helsinki-NLP Opus-MT repository, available on the Hugging Face hub. These models, trained on OPUS parallel corpora, work for low-resource scenarios due to their optimized Transformer architecture, and consistent SentencePiece tokenization—crucial for handling subword segmentation in morphologically rich languages such as Mapudungun.

To study the role of morphological similarity in transfer learning, we use two Spanish-centered parent models: Helsinki-NLP/opus-mt-es-fi, where Finnish is agglutinative and typologically closer to Mapudungun, and Helsinki-NLP/opus-mt-es-en, where English is morphologically simpler and serves as a control. This design enables comparison of transfer effects driven by structural similarity versus general cross-lingual knowledge.

We also explore parent models for Spanish–Turkish, Spanish–Hungarian, and Spanish–French to broaden the typological spectrum. All parent models share similar Transformer architectures, ensuring that differences in transfer performance primarily reflect linguistic rather than architectural factors.

b) Tokenization: Tokenization decomposes text into smaller units to reduce vocabulary size and improve generalization. For morphologically rich languages like Mapudungun, where agglutination produces many rare forms, subword tokenization enables the model to capture reusable morphemes and affixes. We use Byte Pair Encoding (BPE) [18], which iteratively merges frequent character pairs to form subword

units, balancing word- and character-level representations and improving modeling under low-resource conditions.

For the baseline, we use the tokenizer from Duan et al. [1], which is not optimized for Mapudungun morphology. To improve coverage, we build a custom SentencePiece BPE tokenizer trained on monolingual Mapudungun data. Its subword vocabulary is appended to the parent model’s shared vocabulary, enabling better encoding of Mapudungun during fine-tuning. The goal is to produce linguistically meaningful subwords tailored to the specific language.

c) MarianMT Transformer Architecture: Transfer learning experiments use the MarianMT architecture [13], the backbone of Helsinki-NLP Opus-MT models. MarianMT implements the Transformer [16] with a focus on speed, memory efficiency, and modularity, making it suitable for low-resource and multilingual scenarios. It features shared embeddings and vocabulary, with weight tying between source, target, and softmax layers to reduce parameters without degrading performance. A shared SentencePiece vocabulary promotes cross-lingual alignment, which we extend with Mapudungun subwords to preserve morphological integrity. This architecture offers a robust foundation for fine-tuning in low-resource translation.

d) Fine-Tuning Process: Parent models are fine-tuned on 10,000 Spanish–Mapudungun pairs, with 1,250 for validation and testing to match the baseline setup. Training runs for up to three epochs with Adam (weight decay: $\lambda = 0.01$), a fixed learning rate of 3×10^{-4} , and batch size 8, using early stopping based on validation BLEU. Tokenizers are extended with Mapudungun-specific subwords to reduce out-of-vocabulary issues. Fine-tuning, implemented in MarianMT through Hugging Face, completes in under 25 minutes on a single GPU, showing the efficiency of transfer learning in this low-resource setting.

V. RESULTS

A. Overview of Experimental Evaluation

We evaluate transfer learning for low-resource NMT on Spanish–Mapudungun by comparing (i) baseline models trained from scratch and (ii) models fine-tuned from multilingual parent models. Performance is assessed in both directions (arn→es and es→arn) using a fixed dataset split: 10,000 training pairs, 1,250 for validation, and 1,250 for testing. Metrics include BLEU, which measures n-gram precision but is sensitive to word overlap, and chrF, a character-level F-score better suited for morphologically rich languages.

Experiments are structured as follows: 1) Compare baseline and transfer models using a Spanish–Finnish parent, and 2) Assess the effect of parent language morphology by comparing Spanish–Finnish (agglutinative) and Spanish–English (non-agglutinative) parents. Results are reported using the best checkpoint selected by validation BLEU.

B. Impact of Transfer Learning on Translation Quality

a) Mapudungun to Spanish Direction: Table I reports BLEU and chrF scores for (i) a baseline Transformer trained

TABLE I
EFFECT OF TRANSFER LEARNING ON *arn* $\rightarrow$ *es* TRANSLATION.

Model	BLEU Score	chrF Score
Baseline Training	6.26	10.00
Transfer Learning	10.03	30.30

TABLE II
EFFECT OF TRANSFER LEARNING ON SPANISH $\rightarrow$ MAPUDUNGUN
TRANSLATION

Model	BLEU Score	chrF Score
Baseline Training	1.09	10.00
Transfer Learning	9.77	31.20

from scratch (Duan et al., 2020) and (ii) a transfer model fine-tuned from a Spanish–Finnish parent.

As shown in Table I, transfer learning improves performance by +3.77 BLEU and +20.3 chrF over the baseline. The large chrF gain highlights better handling of morphological structure, which is critical for agglutinative languages. These results show that knowledge from high-resource Spanish–Finnish models transfers effectively to Mapudungun–Spanish translation, likely due to shared Spanish structures and encoder representations that adapt during fine-tuning despite typological differences.

b) Spanish to Mapudungun Direction: Table II reports BLEU and chrF scores for the baseline model and the transfer model fine-tuned from a Spanish–Finnish parent.

Transfer learning yields larger gains in this direction, with +8.68 BLEU and +21.2 chrF over the baseline. The baseline’s poor performance reflects the difficulty of generating morphologically complex Mapudungun sequences from limited data. Initializing the decoder from a Spanish–Finnish model appears beneficial, as exposure to morphologically rich Finnish improves the decoder’s ability to handle agglutination. The chrF gain confirms that transfer learning enhances both lexical accuracy and morphological adequacy, showing strong cross-lingual benefits even when parent and target languages are typologically distant.

C. Effect of Parent Language Morphological Similarity

To test whether parent–child morphological similarity affects transfer, we compare Spanish $\rightarrow$ Mapudungun models initialized from: (i) a Spanish–English parent (non-agglutinative) and (ii) a Spanish–Finnish parent (agglutinative and structurally closer to Mapudungun).

Table III presents the results.

Results show no clear advantage for the Finnish-based parent: BLEU scores are nearly identical, and the English-based model slightly outperforms in chrF. This indicates that morphological similarity alone does not drive transfer performance; factors such as vocabulary overlap, tokenizer compatibility, and parent data quality likely play a larger role. Training curves (Figures 2 and 3) further illustrate the evolution of validation BLEU and chrF for both parent models.

TABLE III
EFFECT OF PARENT LANGUAGE ON TRANSFER LEARNING PERFORMANCE
(*es* $\rightarrow$ *arn*)

Parent Language	BLEU Score	chrF Score
English	8.07	29.00
Finnish	8.11	28.00

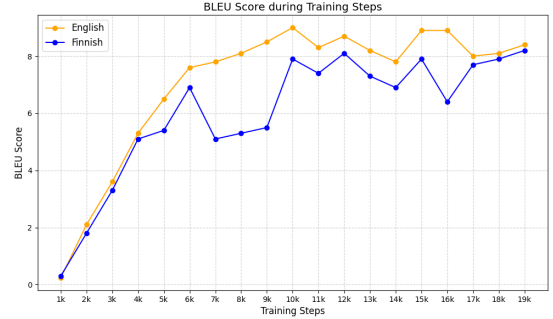


Fig. 2. Validation BLEU score vs. training step for English and Finnish parent models.

Both models converge similarly, plateauing after 15 epochs (18,750 steps). The English-based model shows a slightly faster chrF gain, possibly due to orthographic or lexical similarities with Spanish that improve subword representations. These results suggest that in extremely low-resource settings, tokenizer compatibility and vocabulary integration can outweigh morphological similarity.

D. Tokenization and Vocabulary Analysis

We analyze the differences between the pretrained tokenizer from the Spanish–Finnish parent model and a custom tokenizer trained on monolingual Mapudungun data and merged into the child model’s vocabulary. Consider the following Mapudungun example used in the experiments:

Tüñkülepürakatun (Now I am calmer)

The segmentation of this word by both tokenizers is the following: The Mapudungun word Tüñkülepürakatun is heavily fragmented by the pretrained tokenizer into ["_T", "ü", "nk", "ü", "le", "p", "ü",

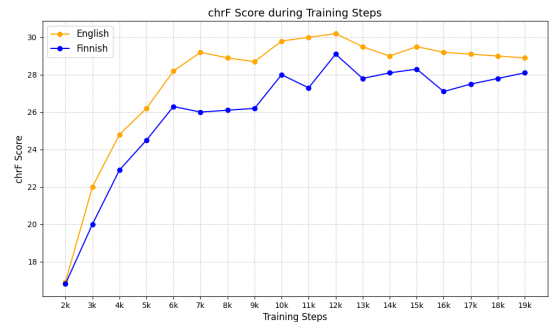


Fig. 3. Validation chrF score vs. training step for English and Finnish parent models.

"ra", "katu", "n"], while the custom tokenizer produces linguistically meaningful units ["_T", "ün", "küle", "püra", "katun"]. The latter segmentation better preserves morphemes and reduces vocabulary sparsity.

The baseline tokenizer breaks words into mostly meaningless character fragments, hindering consistent lexical mappings and training. In contrast, the custom tokenizer captures morphemes such as "küle" ("calm"), "püra" ("more"), and "katun" (verb suffix), producing semantically meaningful units.

Better morphological segmentation improves semantic coherence and eases decoding by reducing the need to reconstruct words from fragments. Integrating Mapudungun subwords into the shared parent vocabulary further aids transfer by aligning embeddings across languages. Text normalization also affects performance: removing diacritics reduces noise but sacrifices linguistic fidelity, while preserving them maintains distinctions. These results show that tokenizer design can influence translation quality as much as, or more than, morphological similarity.

E. Qualitative Analysis of Translations

Automatic metrics like BLEU and chrF capture surface accuracy but overlook semantic fidelity and cultural nuance, which are critical for low-resource, morphologically rich languages. To complement these metrics, we qualitatively compared reference translations with outputs from models initialized with English and Finnish parents.

a) *Example 1: Simple Declarative Sentence:* For the Spanish source "Toda la gente lo ve" ("All the people see it"), the Mapudungun reference is "Kom che pey". The English-initialized model outputs "Kom che pe ngeafuy", closely matching the reference and adding a plausible impersonal/passive suffix (ngeafuy). The Finnish-initialized model instead produces "Che pe pape feyu", a less fluent translation with hallucinated tokens (pape feyu), likely caused by vocabulary mismatches or tokenization errors.

b) *Example 2: Mixed-Code Input and Loanwords:* For the Spanish source "y así nunca más" ("and so never again"), the Mapudungun reference is "y femuechi nunca ma". The Finnish-initialized model outputs "y asi nunca no ma", preserving the mixed Spanish–Mapudungun structure, whereas the English-initialized model produces "fey tu rpu müten", a fully native reconstruction that loses lexical familiarity and semantic nuance. This case highlights how BLEU and chrF fail to capture culturally appropriate variation, as metric-aligned outputs may misrepresent meaning.

F. Error Types and Patterns Observed

Translation errors followed consistent patterns. Overgeneration, more frequent in Finnish-initialized models, added morphemes or words absent in the reference, while undergeneration omitted key elements such as subject markers or negation. Lexical hallucinations, often linked to baseline tokenization, produced rare or non-existent forms. Handling of Spanish loanwords was inconsistent, with models alternately

preserving borrowings or replacing them with native forms depending on vocabulary and tokenizer granularity. These issues indicate that although transfer learning improves quality, core challenges in low-resource NMT persist and may require explicit modeling of code-switching, more contextual decoding, and post-editing constraints.

G. Limitations and Observations

While transfer learning improves performance, several limitations must be considered. The Mapudungun–Spanish corpus required extensive cleaning due to inconsistencies in diacritics, formatting, and annotations; residual noise may still affect results. Diacritic removal simplified training but reduced linguistic precision, potentially obscuring model differences. The lack of linguistic annotation also limits error analysis.

Standard metrics like BLEU and chrF are limited in capturing the complexity of morphologically rich, low-resource languages. BLEU penalizes valid variation, and chrF may miss semantic errors. Both offer only partial insight, underscoring the need for human and typologically informed evaluation.

Finally, native Mapudungun speakers were not involved in development or evaluation. As automatic metrics cannot assess cultural appropriateness, future work should prioritize community engagement to ensure responsible and accurate language technologies.

VI. CONCLUSION

This study shows that transfer learning substantially improves Neural Machine Translation (NMT) for the Spanish–Mapudungun pair, yielding consistent gains in BLEU and chrF scores over models trained from scratch, even with limited data and training. Comparable improvements were observed in both translation directions and for models initialized from Spanish–English and Spanish–Finnish, indicating that morphological similarity between parent and target languages plays a minor role relative to factors such as vocabulary coverage, tokenization, and data quality.

These results highlight three key points: (i) transfer learning is an effective approach for low-resource NMT, (ii) subword modeling and vocabulary alignment may be more critical than structural linguistic similarity, and (iii) automatic metrics alone are insufficient for evaluating translations into culturally rich languages. Future work should explore tokenizer design, longer training schedules, diverse parent languages, and, crucially, active participation of Mapudungun speakers to ensure that translation systems align with community-driven language preservation goals.

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